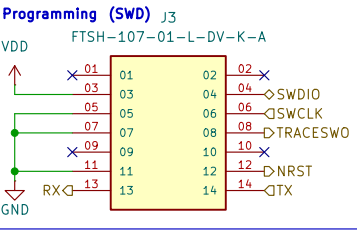


Battery Input

- High-power screw terminal block (Max 65A, 300V)
- Expects either raw wire or terminal connector tabs crimped on the wire across 6–26 AWG
- User will need to hook up an adapter to the connector used by the battery
- Note: 3D model's through-hole pins do NOT match the actual product/footprint (non-issue)
- Add terminal connector crimp kit to BOM

Phase Outputs

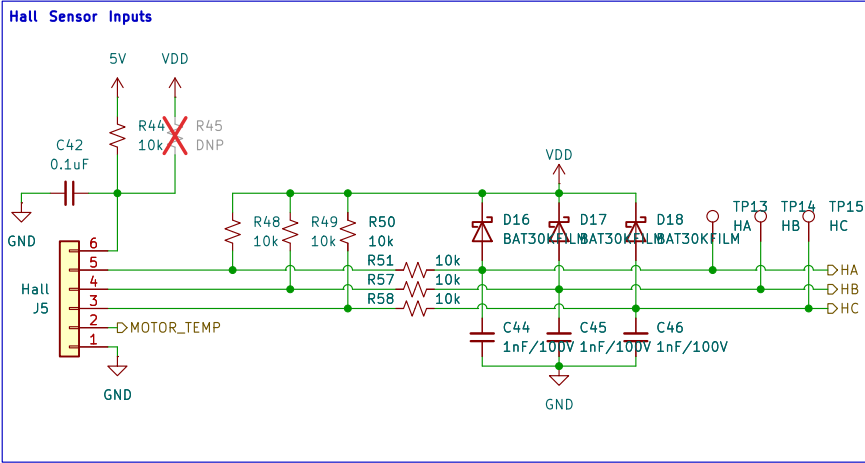
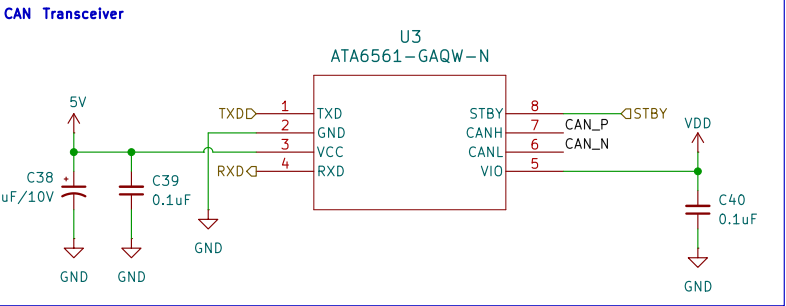
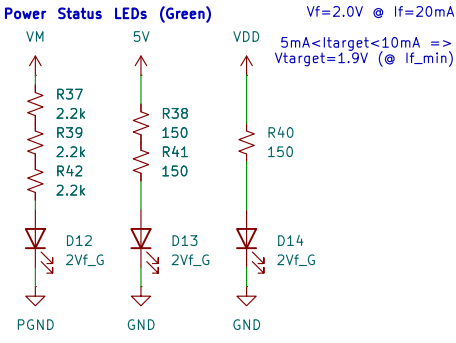
- High-power screw terminal block (Max 65A, 300V)
- Expects either raw wire or terminal connector tabs crimped on the wire across 6–26 AWG



Programming (STDC14)

Expects STLINK-V3, so exposes pins through STDC14 connector

Uses MCU's internal VDD to determine if it's on



Hall Sensor Inputs

Assume VH required by motor hall sensors is 5V, but expose pad to switch to 3V3, if desired (XOR)

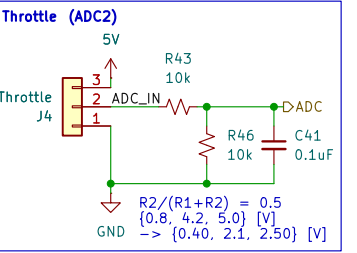
UM2896 5.4: Use CH1–3 of TIM3 (PB4, PB5, PB0) for HA, HB, HC (built-in XOR)

Optional MOTOR_TEMP pin if chosen motor provides winding temps (Note: This is different from the one in Control sheet)

DC12_IN6 on PC0 used for MOTOR_TEMP based on availability

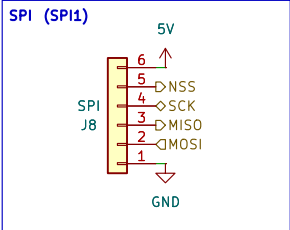
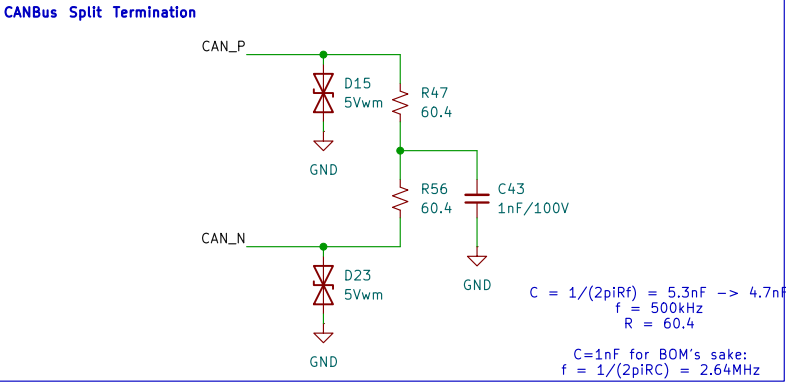
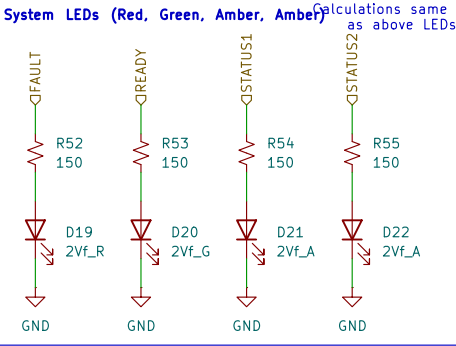
Notes:

- Only open-drain sensors are supported so added pull-up resistors and diodes to clamp voltage at 3V3 to protect MCU
- Active 5V push-pull hall sensors are NOT supported, as the pins couldn't be mapped such that they are 5V-tolerant



Throttle (JST-XH)

- Uses ADC2 or ADC12_IN7 (PC1)
- For bike's thumb throttle (wire cut off & stripped)
- Expects 5V ADC for full range
- Can also use an external pot

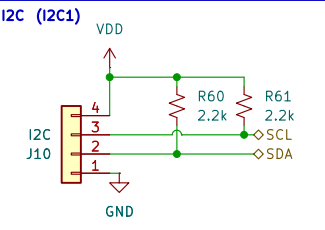


SPI (Optional)

Uses SPI1 (PA4–7), which are NOT 5V-tolerant

Only supports 3.3V TTL

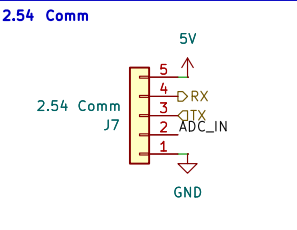
May expect a rotary encoder (alternative to hall sensor)



I2C (Optional)

Uses I2C1_SDA (PB9), I2C1_SCL (PA15)

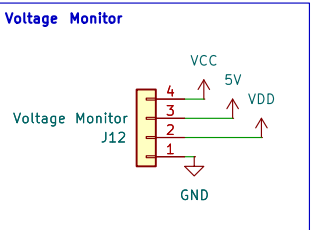
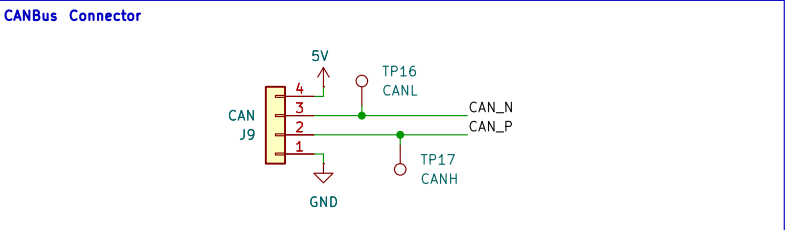
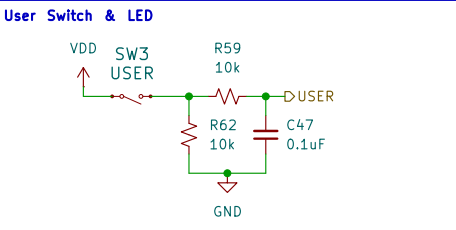
May expect a display? Just exposing it to future-proof



2.54 Comm (2.54mm Pitch Header)

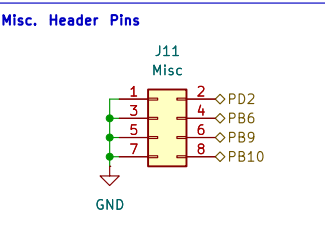
Serial and ADC exposed twice (once with their own JST connectors, and another while combined with regular male pins)

For debugging



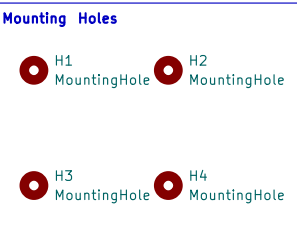
Voltage Monitor

Most power-related signals exposed through test points near pins



Misc. Header Pins

Four unused pins remain, so pair with GND and expose



Mounting Holes

Using non-plated through-holes (NPTH) sized for screw-in rubber bumpers

Add additional parts for standoffs to BOM

Author: Leopoldo Mendoza, Eleanor Wang

Electrium Mobility

Sheet: /Peripherals/

File: peripherals.kicad_sch

Title: ESC 2026 – Peripherals

Size: A3

Date: 2026–08–06

Rev: v1

KiCad E.D.A. 10.0.3

Id: 5/5